

Cameron Beebe, Ph.D.



- **AI:** Numpy, jupyter, conda, keras, deep learning, GPU clusters, Claude Code, LLM APIs, context management, RLMs.
- **Languages:** English (native) and German.
- Technical research and analytical thinking.

🏠 Experience

Dec. 2025– Appy.AI Prompt Engineer for a multi-agent, model-independent harness. Startup. Working with both Product and Engineering teams. Context engineering, harness design, testing. Adjacent research on relevant topics (recursive language models, DSPy, model evals and benchmarks, multi-agent architectures.)

2021– The SciPhi Initiative, LLC. AI research, consulting, and community outreach.

Pair coding and API integration with LLMs in projects like [scipfs](#) and [fungi_fortress](#). Researching capabilities, responsible utilization, and limitations of LLMs.

Leading discussion sessions on AI and society at Nederland Philosophy Club (co-organizer).

Interpreting wave mechanics and quantum probabilities as ideal structural objects: [⟨sciphi⟩_i](#)

2023–2025 Outdoor/Volunteering. Snowboard instructor at Eldora Mountain Resort (2 seasons). Heavy equipment operator, survey reader, data and “math guy” at High Performance Earthworks. Part time gardening and general operations at a cultivation start up (2023-2024). Also volunteer at two other local businesses in Nederland, CO.

2015–2021 PhD at GSN (LMU Munich). AI, cybernetics, and philosophy of science in a structured Ph.D. program at the Graduate School of Systemic Neurosciences.

Dissertation on [Knowledge Transfer in Cognitive Systems Theory](#), looking at transfer learning in artificial neural networks as an example of the effective transfer of control, similar to analogical or model-based knowledge transfer in human cognition. Supported W.R. Ashby’s structural view of systems (cybernetics), and advocated for a model-based view of computation in a world with increasingly diverse computational devices. Took a quantum information course from the physics department, and critiqued quantum-like models of cognition.

Affiliated with the Munich Center for Mathematical Philosophy and the Research Center for Neurophilosophy and Ethics of Neurosciences during this time, and presented work at international conferences.

2019 Deep Learning Intern at Cray, Inc. Worked for Cray’s Distributed Deep Learning Plugin team. Applied, tested, and experimented with the Plugin across multiple GPUs (gradient averaging using MPI), on a variety of architectures including: LSTM, CNN, RL/DQN, CapsNet. Worked mainly with Keras, scheduled jobs via Slurm, managed ML/DL packages and versions with Anaconda and pip, and committed code to internal git repos. (**Python, Jupyter, Slurm, 🐧 Linux, Keras, Anaconda, Bash, Git**).

2013–2015 Open Access: Part time data entry and development for [Open Access LMU](#) database.

Education

- 2021** Ph.D. from [Graduate School of Systemic Neurosciences](#), LMU Munich.
Dissertation: [Knowledge Transfer in Cognitive Systems Theory](#)
Committee: [Stephan Hartmann](#), [Christian Leibold](#), [Ulrike Hahn](#), [Gregory Wheeler](#)
Reviewer: [Peter Asaro](#)
- 2015** Masters in [Logic and Philosophy of Science](#) from [MCMP](#), LMU Munich.
Supervisor: [Karim Thébault](#), Reviewer: [Michael Cuffaro](#)
Thesis: *Fluid Mechanical Models in Physical and Computational Contexts*

Coursework and research in basic formal methods and philosophy of science, including Bayesian networks, probability theory, logic, information theory, and coding ([Netlogo](#)).

Specialization in philosophy of physics.

- 2012** B.A. in Philosophy from Montana State University.
Thesis: *Uncertainty in Quantum Theory* Advisor: [Prasanta Bandyopadhyay](#)

Projects

- 2025 Latest** See latest updates and current projects on [GitHub](#) & [ResearchGate](#)
- 2018 Project/Talk** *Black Boxes or Rube Goldberg Machines? ANNs as Ashby Regulators.* [Epistemic Opacity in Computer Simulation and Machine Learning](#), HLRS. (Related Repo)
- 2018 Project/Talk** Transfer Learning in Artificial Neural Networks: [Introductory Tutorial to Transfer Learning with Keras](#) ([Jupyter](#) Notebook for PyData Meet-up)
- 2014 Project** [Modeling Memory in Signaling Games](#) (Reinforcement learning in [Netlogo](#) w/ Dropout)

Publications

- [Model-Based Computation](#). Natural Computing, Volume 17, Issue 2, p. 271-281. 2018.
- [Sequent Calculus Representations for Quantum Circuits](#). Electronic Proceedings in Theoretical Computer Science, volume 214, p. 3-15. 2016.

WIP Manuscripts (More on my [ResearchGate Profile](#).)

- [Artificial Neural Networks as Cybernetic Regulators](#). [Bayesian Confirmation from Analog Models](#). With [Roland Poellinger](#). [UNDER REVIEW]
- [Transfer Learning: The Logic of Discovery for Artificial General Intelligence](#) [Decoherence and Survival: Why Incompatible Observables Should be Compatible](#).